Section A

Answer **all** the questions in this section.

1 Fig. 1.1 is an electron micrograph of part of a eukaryotic cell with organelles involved in protein synthesis.

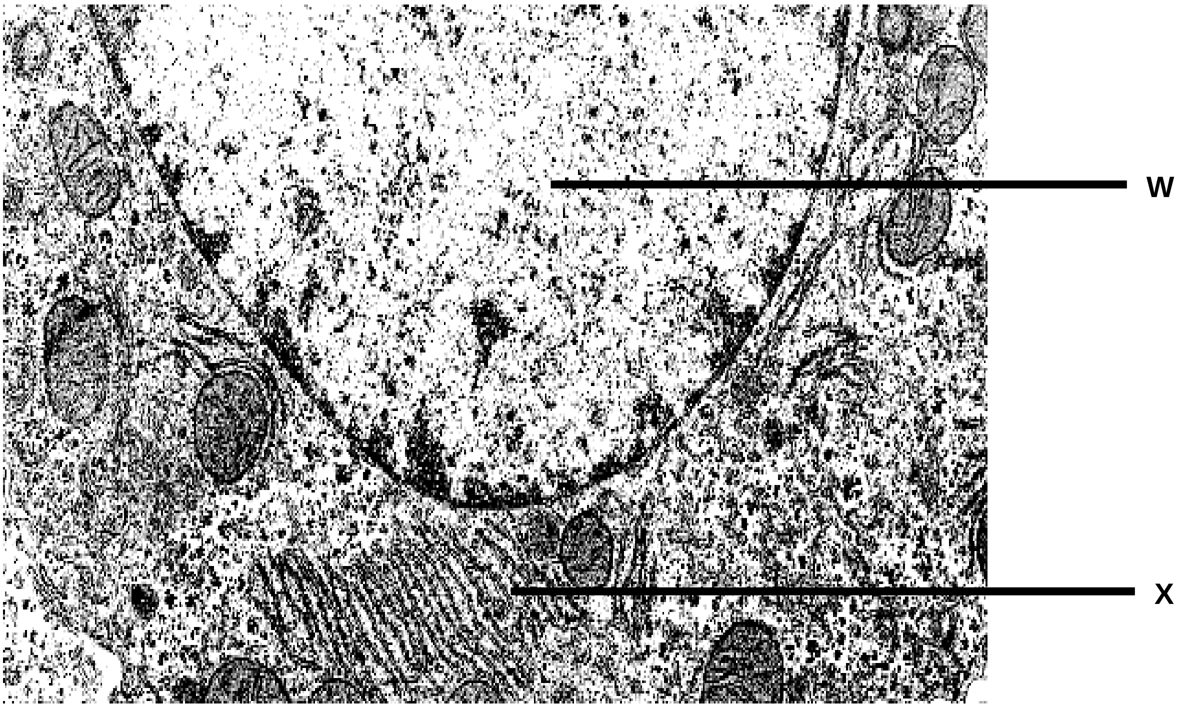

Fig. 1.1

(a) State **three** ways in which Organelle **W** differs from Organelle **X**. [3]

An experiment was carried out to determine what happens to amino acids after they are absorbed by animal cells. The cells were incubated for 5 minutes in a medium containing radioactively labelled amino acids. The radioactive amino acids were then washed off and the cells were incubated in a medium containing only non-radioactive amino acids.

Samples of the cells were removed from the medium every five minutes for 40 minutes. For each sample, the levels of radioactivity in three different organelles, **P**, **Q** and **R**, were determined. The results of the experiment are shown in Fig. 1.2.

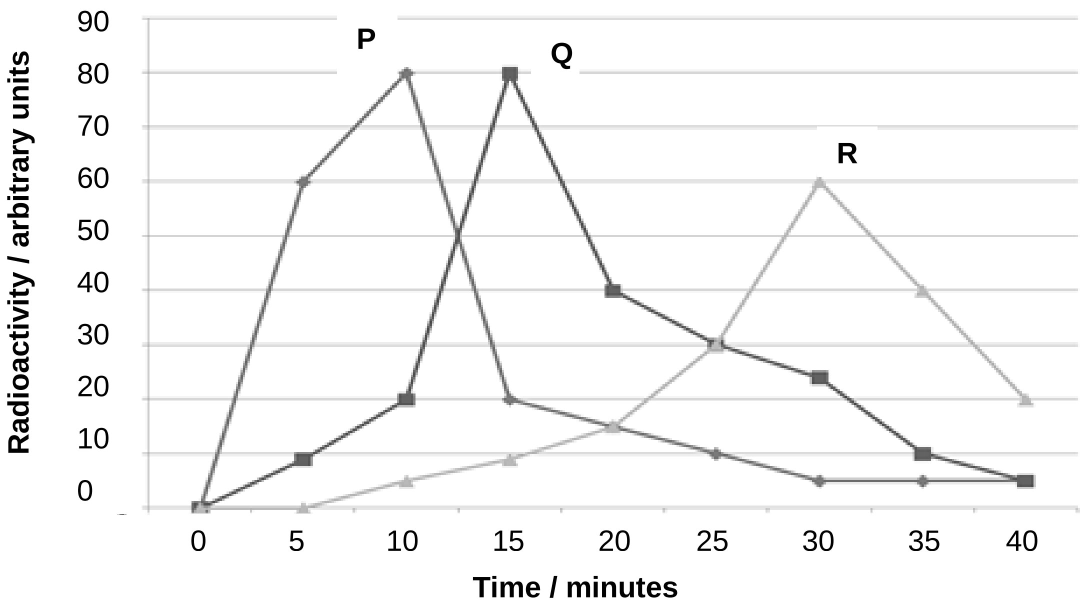

Fig. 1.2

(b) With reference to Fig. 1.2, explain the changes in the level of radioactivity of the three organelles with time. [3]

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Fig. 1.3 shows the central dogma of molecular biology.

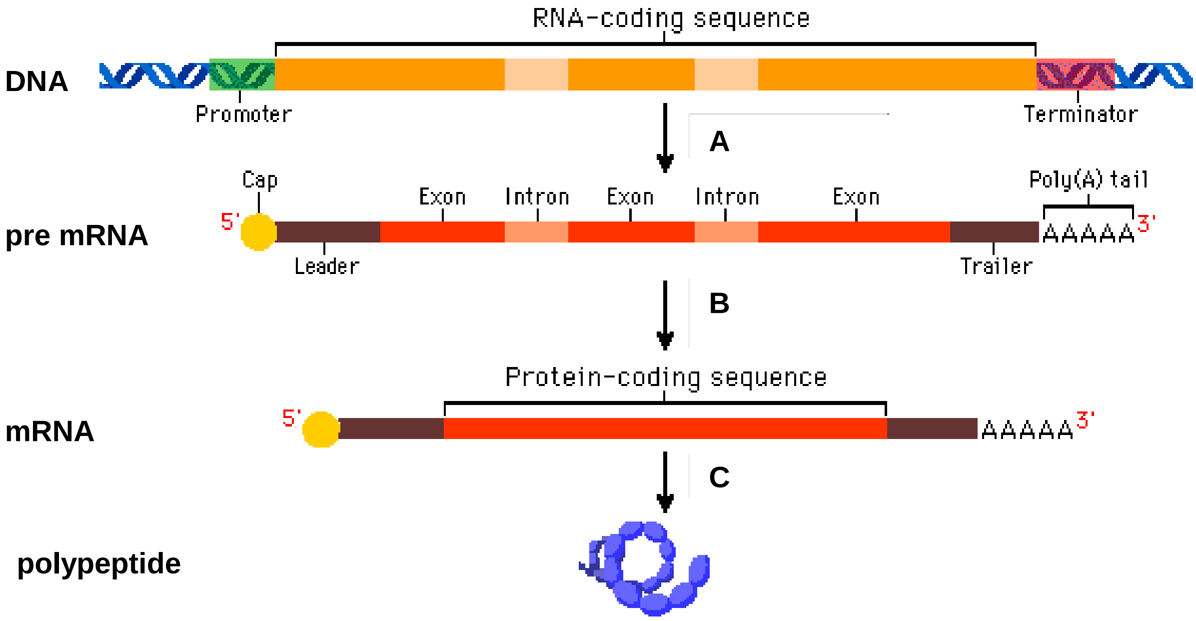

Fig. 1.3

(c) **(i)** With reference to Fig. 1.3, explain how the pre mRNA is processed to form mature mRNA for translation. [3]

(ii) Transfer RNA (tRNA) is one of the molecules involved in translation. Describe one structural feature which adapts tRNA to its role in translation. [1]

[Total: 10]

2 In some species of the locusts, the female has a diploid chromosome number of 16 while the male locust has a diploid number of 15. The difference in chromosome number between the sexes is the result of a difference in the number of sex chromosomes (X chromosome). Female locusts have two X chromosomes (XX) and male locusts have only one X chromosome and no other sex chromosome (XO).

(a) Explain what is meant by the *"diploid chromosome number"*. [1]

(b) Explain why there is a need for reduction division prior to fertilisation in sexual reproduction. [2]

(c) At the end of meiosis II, state the number of chromosomes that will be present in the gametes of:

(i) a male locust; [1]

(ii) a female locust. [1]

(d) Darwinism is a theory of evolution based upon inherited variations in organisms and natural selection of fitter variants to produce species adapted to their habitats.

Twentieth-century biology added a theory of inheritance, the science of genetics, to give Neo-Darwinism. In the past 20 years the techniques of genetics and molecular biology have converged to provide both a remarkably detailed understanding of the genes that define the molecular composition of any organism and the ability to transfer genes from one species to another.

(i) With reference to the information above, explain how molecular evidence derived from techniques of genetics and molecular biology supports Darwin's theory of natural selection. [3]

(ii) One important aspect of the modern theory of evolution by natural selection is the presence of genetic variation in a natural population.

Describe the role of meiosis in natural selection. [2]

[Total: 10]

3 In cats, the gene controlling fur coat colour is sex-linked. This gene has two alleles. In a particular colony of cats, the coats of the males were either black or ginger and those of the females, black, ginger or tortoiseshell (a patchwork black, ginger and cream).

Fig. 3.1 shows a picture of a tortoiseshell cat.

Fig. 3.1

(a) With reference to the information above, state the genotypes according to their corresponding phenotypes of the different coats of the male cats using appropriate symbols. [1]

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(b) Using a genetic diagram, determine the phenotypes expected in the progeny of a cross between a black female and a ginger male. [4]

(c) In a litter of kittens, half the number of females was tortoiseshell and the other half was black; half the number of males was black and the other half was ginger.

(i) State the colours of the parental male and female cats. [2]

Male ______________________________

Female ______________________________

(ii) Explain your answers in **(c)(i)**. [1]

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(d) A research team has also identified that external factors can also play a role in influencing the fur coat colour of the cats.

Describe how a named external factor can influence the fur coat colours of cats. [2]

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[Total: 10]

4 Restriction enzymes cut DNA molecules only at specific sites with particular base sequences. The target sites for the enzymes **A**, **B** and **C** are shown in Table 4.1 below. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

Table 4.1

Name of restriction enzyme	DNA base sequence of target site
A	-C-C-C G-G-G- -G-G-G C-C-C-
B	-C- C-G-G- -G-G-C- C-
C	-A -A-G-G-T-T- -T-T-C-G-A- A-

The target base sequences for all three restriction enzymes in Table 4.1 are palindromic.

(a) **(i)** With reference to Table 4.1, explain what is meant by the term *palindromic*. [1]

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(ii) Suggest why it is preferable that restriction enzyme target sequences be palindromic. [1]

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Restriction digests were performed separately on pure samples of a plasmid using four different restriction enzymes, RE1, RE2, RE3 and RE4.

Lanes 1, 2, 3 and 4 of Fig. 4.2 show the results of the restriction digests using restriction enzymes, RE1, RE2, RE3 and RE4 respectively. A DNA ladder (with known fragment lengths) was loaded in Lane **5**.

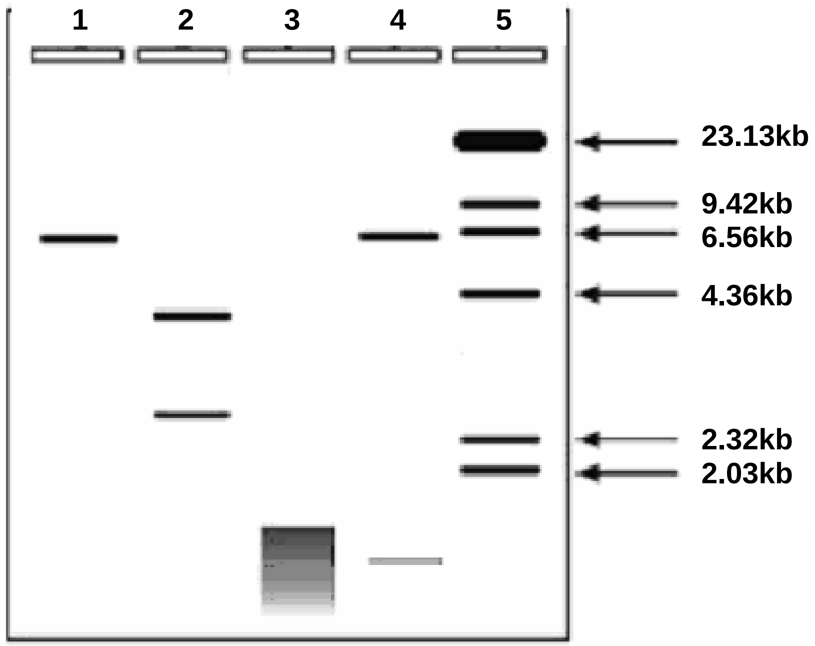

Fig. 4.2

(b) **(i)** Using Fig. 4.2, identify the approximate size of the plasmid used in the restriction digests. [1]

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(ii) State **one** limitation in using a plasmid vector. [1]

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(iii) If this experiment was performed without any error, suggest a reason why a smear was obtained in lane **3**. [2]

(iv) Polymerase chain reaction (PCR) was performed to amplify the quantity of the plasmid used in restriction digest.

Describe the roles of the two forward and reverse primers in PCR. [2]

(c) The plasmid was found to only contain one selectable marker.

Suggest why a plasmid containing one selectable marker is not ideal for use in cloning when we want to insert a foreign gene into *E.coli*. [2]

[Total: 10]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where applicable.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

5 **(a)** Glyceraldehyde 3-phosphate is the precursor to forming glucose during the Calvin Cycle. Glucose is polymerised into starch, a macromolecule for storage.

Outline how glyceraldehyde 3-phosphate is made in the Calvin cycle and converted into starch for storage. [6]

(b) There are many factors that can affect the rate of photosynthesis in plants.

With reference to the term 'limiting factor', explain how two named factors can have effects on photosynthesis. [8]

(c) It might be possible to produce genetically modified plants with increased photosynthetic enzymes with enhanced efficiency.

Discuss the ethical implications of genetically modified plants. [6]

[Total: 20]

Or

6 **(a)** The link reaction is an important process between glycolysis and the Krebs cycle in aerobic respiration.

Outline the link reaction and state where it occurs. [6]

(b) Some plants will die from alcohol accumulation when submerged in water for a prolonged period of time. Athletes may accumulate lactic acid in their muscles when they engaged in strenuous activity.

Explain how plants and humans accumulate alcohol and lactic acid respectively. [8]

(c) Oxidative phosphorylation produces a total of 28 ATP from re-oxidising NADH and $FADH_2$.

Discuss the roles of NADH and $FADH_2$ in aerobic respiration. [6]

[Total: 20]

– End of paper –